

Test Results, Observations and Use Case Summary – Brute Force Detection and Automated Response

1. Use Case Objective

The objective of this use case was to detect repeated failed login attempts on a Windows system using Splunk Enterprise and automatically respond through Splunk SOAR.

The solution was designed to identify potential brute force attacks by monitoring Windows Security Event ID 4625 (Failed Logon Events). When a predefined threshold of failed login attempts was exceeded, Splunk generated an alert and forwarded the event to Splunk SOAR for automated response.

2. Test Environment

SIEM Components

- Splunk Enterprise Search Head
- Splunk Indexer
- Splunk Heavy Forwarder

SOAR Components

- Splunk SOAR 7.1.0.225

Data Sources

- Windows Server 2022
- Windows Client

Cloud Platform

- Amazon Web Services (AWS EC2)

3. Test Scenario

The following attack scenario was simulated:

1. Multiple failed login attempts were generated against a Windows user account.
2. Windows logged Event ID 4625 events.
3. Splunk Universal Forwarder collected and forwarded logs.
4. Splunk alert logic detected the attack.
5. Alert was forwarded to Splunk SOAR.
6. SOAR playbook executed automated response actions.
7. User account was disabled automatically.

4. Test Results

Test Step	Expected Result	Actual Result	Status
Failed login attempts generated	Windows logs Event ID 4625	Event captured successfully	Pass
Logs forwarded to Splunk	Events visible in Splunk Search	Events received successfully	Pass
Detection SPL executed	Alert triggered after threshold exceeded	Alert triggered successfully	Pass
Alert forwarded to SOAR	Event received by SOAR	Event received successfully	Pass
SOAR playbook executed	Automated workflow runs	Workflow executed successfully	Pass
User account disabled	User disabled in Active Directory	User disabled successfully	Pass

5. Key Observations

Observation 1

Windows Event ID 4625 provides reliable visibility into failed authentication attempts.

Observation 2

Threshold-based detection significantly reduces false positives compared to alerting on individual failed logins.

Observation 3

Automated response through SOAR reduces analyst response time and helps contain potential threats quickly.

Observation 4

Integration between Splunk Enterprise and Splunk SOAR enables end-to-end detection and response workflows.

6. Security Benefits

- Early detection of brute force attacks
- Automated incident response
- Reduced manual effort for SOC analysts
- Faster containment of compromised accounts
- Improved security monitoring visibility

7. Potential False Positives

The detection logic may generate alerts during legitimate scenarios such as:

- Users repeatedly entering incorrect passwords.
- Cached credentials using outdated passwords.
- Misconfigured service accounts.
- Application authentication failures.

Analysts should review authentication logs and user activity before classifying the event as malicious.

8. Challenges Encountered

- Initial tuning of alert thresholds
- Validation of SOAR integration
- Testing account disable actions safely in a lab environment
- Preventing duplicate alerts during testing

9. Conclusion

This use case successfully demonstrated an end-to-end Security Operations workflow using Splunk Enterprise and Splunk SOAR. The solution detected brute force activity, generated alerts, automatically initiated response actions, and disabled the targeted account without manual intervention.

The implementation showcases SIEM monitoring, alert engineering, SOAR automation, incident response, and security operations concepts commonly used in modern SOC environments.